General Matrix Multiplication (Optional)

This section provides the general rule for matrix multiplication, which is the core operation for vectorizing a neural network.

The Transpose Operation (A^T)

- A **matrix** is a 2D array of numbers, defined by its `(rows, columns)`.
- The **transpose** of a matrix, A^T, is found by taking the **columns** of A and turning them into the **rows** of A^T.
- **Example:** If A is a **2x3** matrix (2 rows, 3 cols), then A^T will be a **3x2** matrix (3 rows, 2 cols).

The Rule for Matrix Multiplication: $Z = A \times W$ 🧮

The core rule for multiplying two matrices (e.g., A^T and W) is to compute the dot product of rows from the first matrix with columns from the second.

- To find the element in **row** i and **column** j of the output matrix Z, you take the dot product of **row i of the first matrix** and **column j of the second matrix**.
- $Z(i, j) = (\text{Row } i \text{ of } A^T) \cdot (\text{Column } j \text{ of } W)$

Example Calculation

Let's calculate a few elements of $Z = A^T \times W$.

- **To find $Z(1, 1)$ (Row 1, Col 1):**
  - $(\text{Row } 1 \text{ of } A^T) \cdot (\text{Col } 1 \text{ of } W)$
  - $[1, 2] \cdot [3, 4] = (1 \times 3) + (2 \times 4) = \mathbf{11}$
- **To find $Z(3, 2)$ (Row 3, Col 2):**
  - $(\text{Row } 3 \text{ of } A^T) \cdot (\text{Col } 2 \text{ of } W)$
  - $[0.1, 0.2] \cdot [5, 6] = (0.1 \times 5) + (0.2 \times 6) = 0.5 + 1.2 = \mathbf{1.7}$
- **To find $Z(2, 3)$ (Row 2, Col 3):**
  - $(\text{Row } 2 \text{ of } A^T) \cdot (\text{Col } 3 \text{ of } W)$
  - $[-1, -2] \cdot [7, 8] = (-1 \times 7) + (-2 \times 8) = -7 - 16 = \mathbf{-23}$

This process is repeated for every element to compute the full output matrix Z.

Dimension Rules for Multiplication 📏

You cannot multiply matrices of any arbitrary size. They must follow two specific rules.

- **1. Requirement:** To multiply `Matrix1` × `Matrix2`, the number of **columns** in `Matrix1` must **equal** the number of **rows** in `Matrix2`.
  - *Example:* A^T (a **3x2** matrix) $\times$ W (a **2x4** matrix).
  - The "inner dimensions" (2 and 2) match, so the operation is valid.
  - This is required because the dot product must be between vectors of the same length.
- **2. Output Shape:** The resulting matrix, Z, will have the "outer dimensions."
  - *Shape:* (Number of **rows** from `Matrix1`) × (Number of **columns** from `Matrix2`).
  - *Example:* Z will be a **3x4** matrix.

Key Takeaway

This matrix multiplication rule is the key to **vectorization**. It allows a neural network to perform all the dot products for an entire layer in one highly efficient operation, which is why `np.matmul` runs so much faster than a `for` loop.